🧠 NCLEX STROKE (CEREBROVASCULAR ACCIDENT - CVA) OVERVIEW

1. WHAT IS A STROKE?

- **A stroke (CVA - cerebrovascular accident) happens when blood flow to the brain is blocked or reduced, causing brain cells to die.**
- **Stroke is a medical emergency! The faster treatment is given, the better the recovery.**
- **Types of Stroke:**
  - 🔹 **Ischemic Stroke** (Most common - caused by a blood clot).
  - 🔹 **Hemorrhagic Stroke** (Caused by bleeding in the brain).
  - 🔹 **Transient Ischemic Attack (TIA - "Mini-Stroke")** (Temporary blockage, no permanent damage).

2. TYPES OF STROKE & CAUSES

Type of Stroke	Definition (What Happens?)	Main Cause (Why It Happens?)
Ischemic Stroke (87% of strokes)	Blood clot blocks blood flow to the brain.	Atherosclerosis (fat buildup in arteries), blood clot from heart (Atrial Fibrillation - irregular heartbeat).
Hemorrhagic Stroke (13% of strokes)	Blood vessel in the brain bursts, causing bleeding.	High blood pressure (hypertension), aneurysm (weakened blood vessel that bursts), trauma (head injury).
Transient Ischemic Attack (TIA - "Mini-Stroke")	Temporary blockage of blood flow to the brain (lasts minutes to hours, no permanent damage).	Small blood clots that dissolve, warning sign of future stroke.

3. STROKE WARNING SIGNS (FAST MNEMONIC)

🆘 **Think FAST! Call 911 IMMEDIATELY if you see these signs!**

- **F – Face Drooping** (One side of the face looks uneven or droops).
- **A – Arm Weakness** (One arm drifts downward when raised).
- **S – Speech Difficulty** (Slurred, slow, or confused speech).
- **T – Time to Call 911** (Get help IMMEDIATELY!).

🚨 **Other Symptoms:**
✅ **Sudden numbness or weakness (especially on one side of the body).**
✅ **Vision problems (blurry or loss of vision in one or both eyes).**
✅ **Severe headache (especially for hemorrhagic stroke).**
✅ **Loss of balance, dizziness, difficulty walking.**

4. RISK FACTORS FOR STROKE

MODIFIABLE RISK FACTORS (CAN BE CHANGED WITH LIFESTYLE CHANGES & TREATMENT)

❌ **High Blood Pressure (Hypertension - #1 Cause!).**
❌ **Smoking (Increases blood clot risk).**
❌ **Diabetes (High blood sugar damages blood vessels).**
❌ **High Cholesterol (Leads to clogged arteries).**
❌ **Obesity (Extra weight increases stroke risk).**
❌ **Atrial Fibrillation (AFib - Irregular heart rhythm that causes blood clots).**
❌ **Alcohol & Drug Use (Heavy drinking & cocaine use increase stroke risk).**

NON-MODIFIABLE RISK FACTORS (CANNOT BE CHANGED)

- **Age > 55 (Risk increases with age).**
- **Family History of Stroke.**

- 🔹 Race/Ethnicity (Higher risk in African Americans & Hispanics).
- 🔹 Previous TIA (Warning sign for a future stroke!).

5. DIAGNOSIS OF STROKE

✅ CT Scan (Computed Tomography) – First test to check for bleeding.
✅ MRI (Magnetic Resonance Imaging) – Shows brain damage from stroke.
✅ Carotid Ultrasound – Checks for blocked arteries in the neck.
✅ Echocardiogram – Looks for blood clots in the heart (especially in AFib patients).
✅ Blood Tests – Checks for clotting problems, blood sugar, cholesterol.

6. TREATMENT OF STROKE

A. ISCHEMIC STROKE (CLOT STROKE - MOST COMMON TYPE)

✅ Tissue Plasminogen Activator (tPA – Clot-Busting Drug)

- Breaks up blood clots if given within 3-4.5 hours of stroke onset!
- 🚨 Do NOT give if bleeding is present (hemorrhagic stroke).
  ✅ Aspirin or Clopidogrel (Blood Thinners) – Prevent future clots.
  ✅ Mechanical Thrombectomy (Clot Removal Surgery) – Used for large clots.

B. HEMORRHAGIC STROKE (BLEEDING STROKE)

✅ Lower Blood Pressure (To prevent further bleeding).
✅ Surgery (To repair aneurysm or remove excess blood).
✅ Stop Blood Thinners (Like Warfarin, Aspirin – Can make bleeding worse!).

C. TIA (TRANSIENT ISCHEMIC ATTACK - MINI-STROKE)

✅ Blood Thinners (Aspirin, Clopidogrel – Prevent future stroke).
✅ Control High Blood Pressure, Diabetes, & Cholesterol.

7. STROKE REHABILITATION & RECOVERY

✅ Physical Therapy (Helps regain movement & strength).
✅ Occupational Therapy (Helps with daily tasks like dressing & eating).
✅ Speech Therapy (Helps with speaking, swallowing, & memory problems).
✅ Emotional Support (Stroke survivors may have depression & frustration).

8. STROKE COMPLICATIONS

⚠ Paralysis or Weakness on One Side (Hemiparesis).
⚠ Difficulty Speaking (Aphasia).
⚠ Swallowing Problems (Dysphagia - Risk for choking & aspiration pneumonia).
⚠ Memory & Thinking Problems (Cognitive Impairment).
⚠ Depression & Mood Changes.

🚨 Nursing Priority: Assess for Swallowing Problems (Dysphagia) Before Giving Food or Drinks!

- Use a Swallow Test (Speech Therapist may do further testing).
- Prevent Choking & Aspiration Pneumonia!

9. STROKE PREVENTION STRATEGIES

✅ Control High Blood Pressure (Goal: < 130/80 mmHg).
✅ Quit Smoking & Avoid Excessive Alcohol.
✅ Exercise Regularly (At Least 30 Minutes, 5 Days a Week).
✅ Eat a Healthy Diet (Low in Salt, Fat, & Cholesterol).
✅ Take Prescribed Blood Thinners (If At Risk).

10. NCLEX "MUST-KNOW" STROKE CONCEPTS

- 🔹 **FAST Mnemonic = Best Way to Recognize a Stroke!**
- 🔹 **tPA (Clot-Busting Drug) Must Be Given Within 3-4.5 Hours!**
- 🔹 **Ischemic Stroke = Clot Blockage, Hemorrhagic Stroke = Bleeding!**
- 🔹 **CT Scan = First Test to Determine Stroke Type!**
- 🔹 **High Blood Pressure = #1 Risk Factor for Stroke!**
- 🔹 **Swallow Test Required Before Giving Food (Prevent Choking & Aspiration Pneumonia)!**
- 🔹 **Left Brain Stroke = Speech & Language Problems!**
- 🔹 **Right Brain Stroke = Poor Judgment & Vision Problems!**